Vegetation Indices

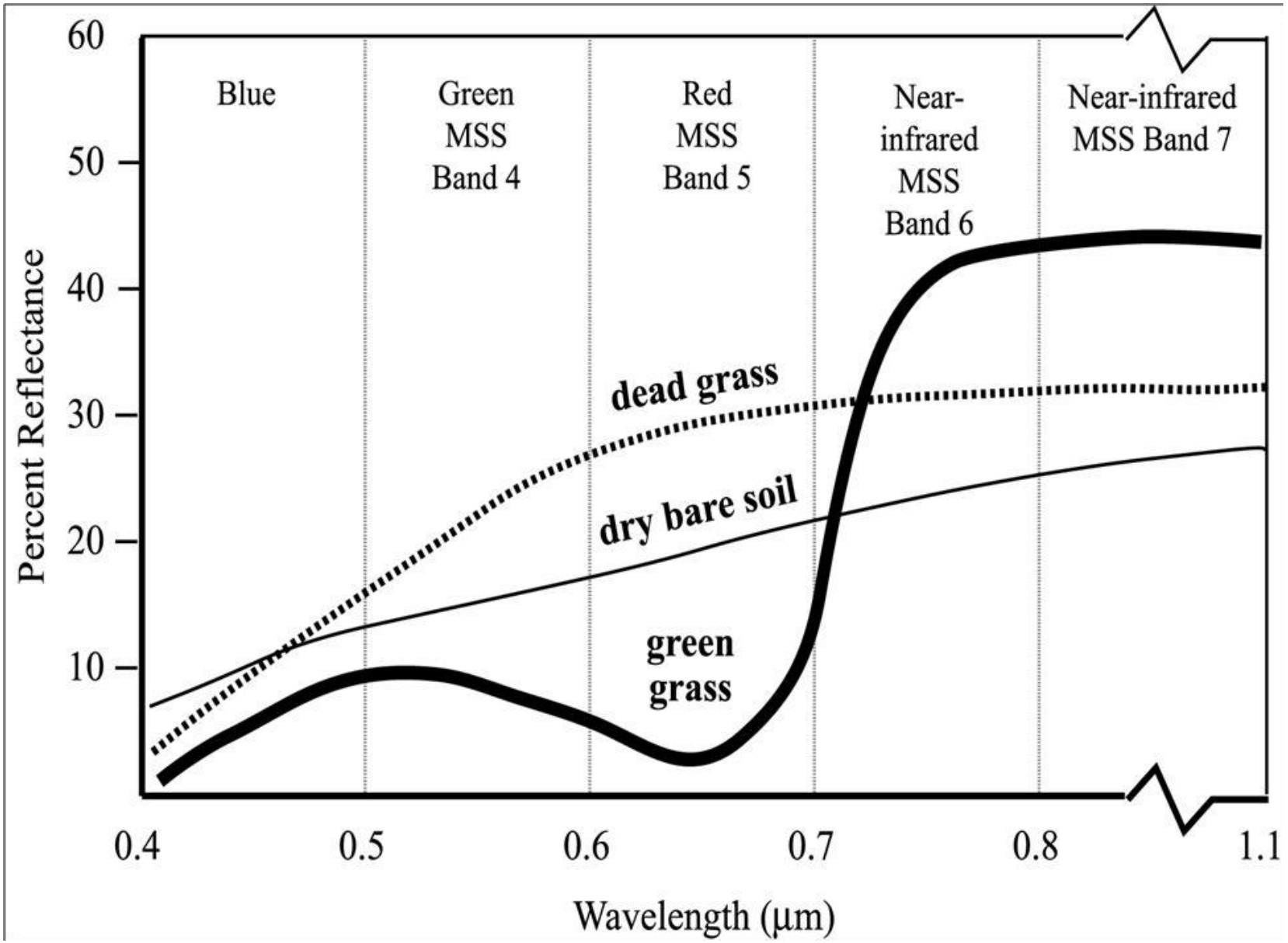
Vegetation Spectral Response

- Natural surfaces are about equally bright in the red and near-infrared. An exception is green vegetation
- Red light is strongly absorbed by chlorophyll while Near-infrared light is highly reflected by the same
- Area with much green vegetation will be very bright in the near-infrared and very dark in the red part spectrum

Vegetation Spectral Response

- Green vegetation has a distinctive reflectance curve:
  - Absorption in blue
  - Reflection in green
  - Absorption in red
  - Strong reflection in NIR
- Thus, the discrimination of vegetation from other surface cover types is significantly enhanced by exploiting red and IR wavelength range

Reflectance Curves for Selected Phenomena



Vegetation Spectral Response

NIR reflectance of vegetation will change

- With stress
- Disease
- Water stress
- Old age

Vegetation Indices

What are they and what do they do...?

- A Vegetation Index (VI) is an "image layer" created from the existing bands of a multispectral image
- This new layer often provides unique and valuable information not found in any of the other individual bands

Vegetation Indices

- Formed from combination of two to several spectral bands that are added, divided, multiplied to yield single value that indicates the amount or vigor of vegetation with in a pixel
- Function of the two channels to enhance response to vegetation
- Quantitative measures for vegetation abundance and vigor

Vegetation Indices

- Enables to identify areas of unhealthy or stressed vegetation
- Significantly enhance vegetation from other surface cover types, thus helps in the discrimination
- Stressed vegetation show low near-infrared reflectance, so the ratios would be lower than for healthy green vegetation
- Tracking of temporal characteristics/sensitivity of vegetation

Vegetation Indices

- Can reduce sensitivity to
  - Topographic Effects
  - Minimize response to unnecessary factors, Background (soil)
  - View/illumination angle
  - Atmosphere

Vegetation Indices

- Main categories:
  - Ratio indices
  - Perpendicular indices

– Ratio Indices

  - Ratio Vegetation Index (RVI)
  - Normalized Difference Vegetation Index (NDVI)

– Perpendicular Indices

  - Perpendicular Vegetation Index (PVI)
  - Soil Adjusted Vegetation Index (SAVI)

Different Vegetation Indexes

- More complex ratios involving the sums of and differences between spectral bands for various sensors, have been developed for monitoring vegetation conditions
- **Different Vegetation Indexes**
- RVI-Ratio Vegetation Index
- DVI-Difference Vegetation Index
- NDVI-Normalized Difference Vegetation Index
- PVI-Perpendicular Vegetation Index
- IPVI-Infrared Percentage Vegetation Index
- SAVI-Soil Adjust Vegetation Index
- MSAVI-Modified Soil Adjusted Vegetation Index
- GVI-Green Vegetation Index

and many others

Image ratios and indices

- **Ratios**
  - Red / Mid-IR … to identify snow / ice (ref- Lec spectral reflectance)

- **Ratio Vegetation Index (RVI)** = **Near IR / Red** …… if < 1 = un-vegetated

* RVI can create infinite values

- **Difference Vegetation Index (DVI)** = **NIR - Red** ……if < 0 = un-vegetated

Infrared/Red Ratio Vegetation Index

- The near-infrared (NIR) to red simple ratio (SR) is the first true vegetation index:
- $SR = \frac{\boldsymbol{NIR}}{\boldsymbol{RED}}$ $RVI = \frac{NIR}{RED}$ $RVI = \frac{\rho_{NIR}}{\rho_{RED}}$
- It takes advantage of the inverse relationship between chlorophyll absorption of red radiant energy and increased reflectance of near-infrared energy for healthy plant canopies
- RVI itself is no longer generally used in remote sensing. Instead normalized difference vegetation index (NDVI) is used

Normalized Difference Vegetation Index

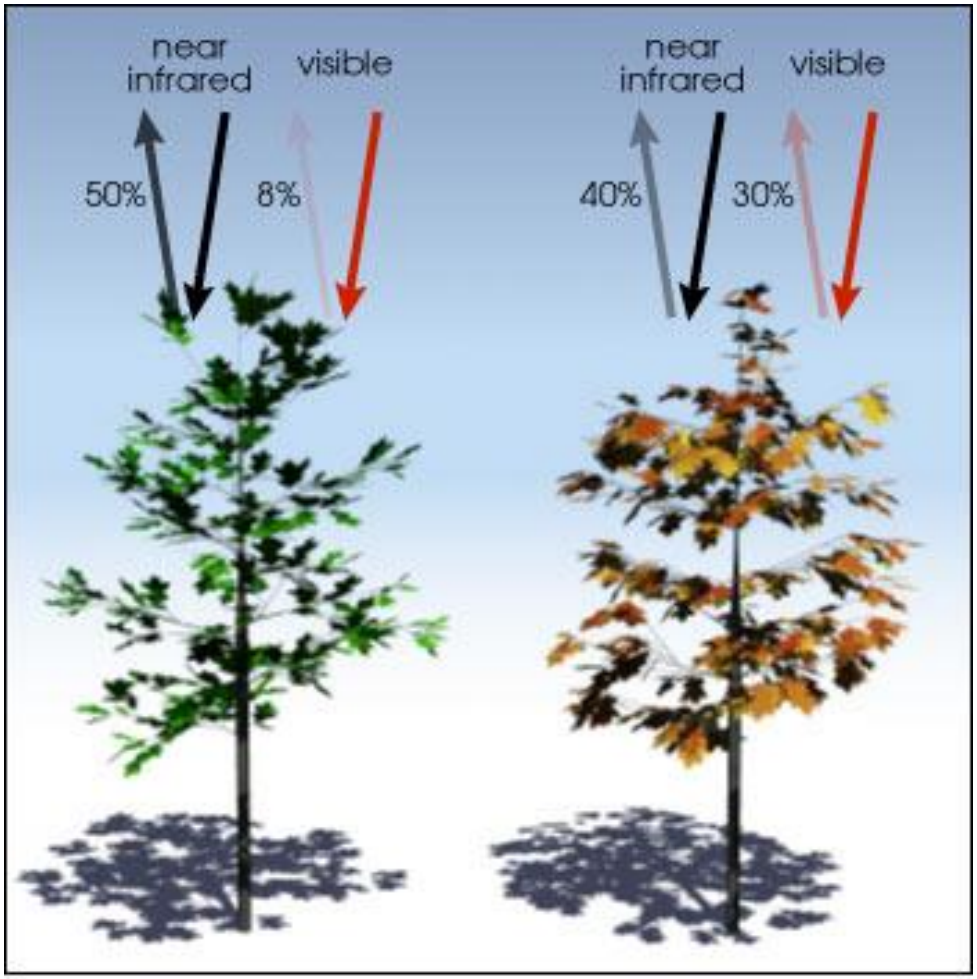
- The generic normalized difference vegetation index (NDVI):

$$NDVI = \frac{NIR - RED}{NIR + RED} \qquad NDVI = \frac{\rho_{NIR} - \rho_{RED}}{\rho_{NIR} + \rho_{RED}}$$

- ρ = Reflectance

yields values between -1 and 1

- Negative values of NDVI (values approaching -1) correspond to water
- Values close to zero (-0.1 to 0.1) = barren areas of rock, sand, or snow
- Low positive values represent shrub and grassland
- High values indicate very healthy vegetation



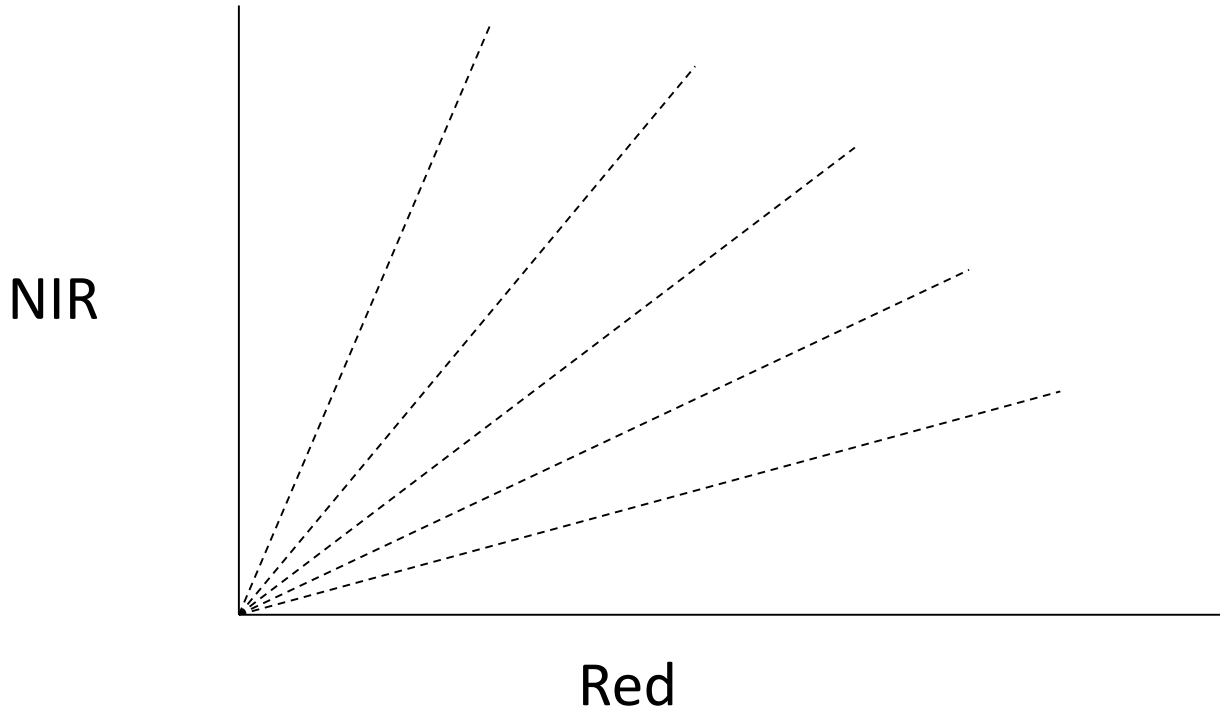
$$\frac{(0.50 - 0.08)}{(0.50 + 0.08)} = 0.72$$

$$\frac{(0.4 - 0.30)}{(0.4 + 0.30)} = 0.14$$

NDVI

(courtesy http://earthobservatory.nasa.gov)

- Both RVI and NDVI basically measure the slope of the line between the origin of red-NIR space and the red-NIR value of the image pixel



Vegetation Indices

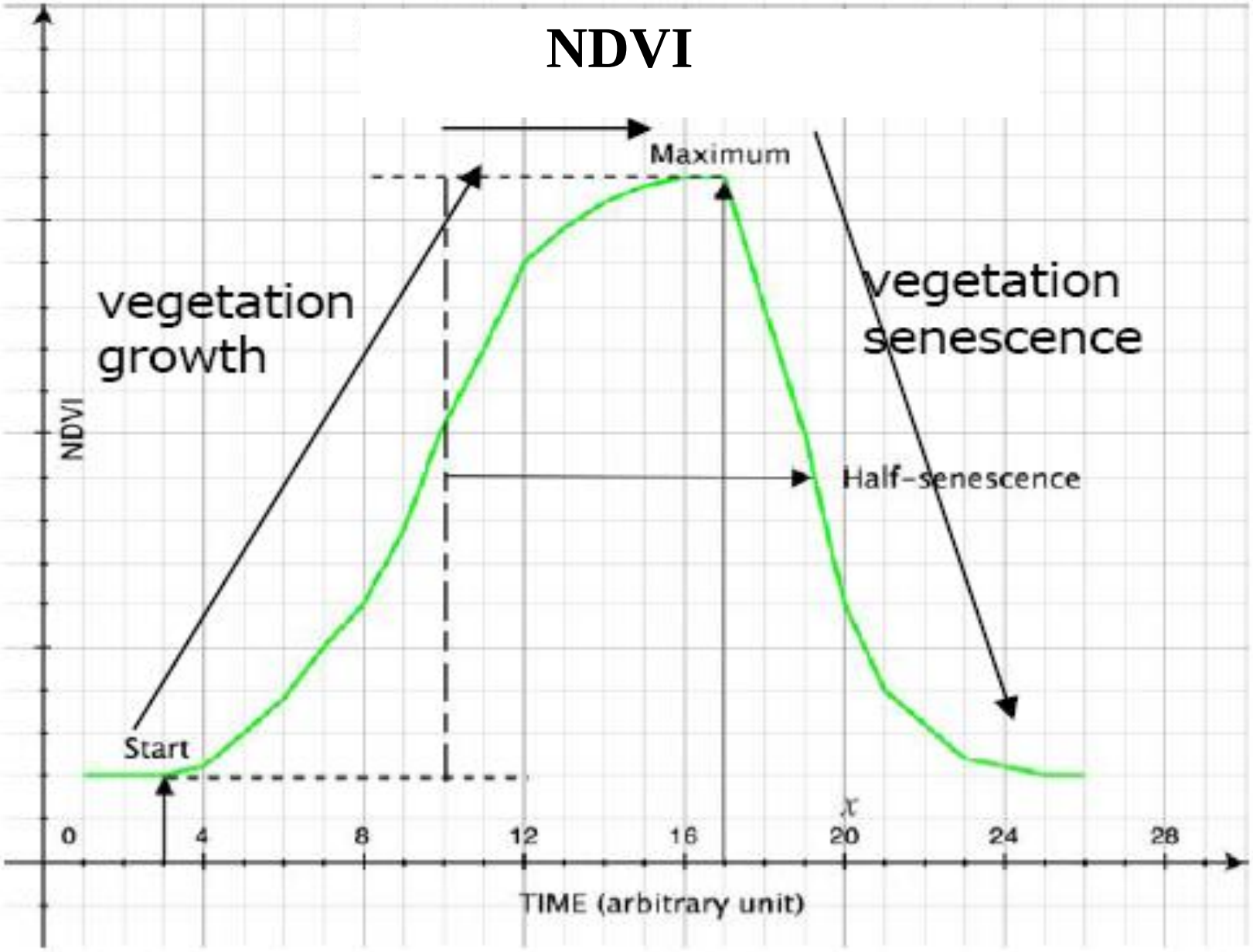
- **Normalized Difference Vegetation Index (NDVI) is** one of the widely used image transform
- It is used to monitor vegetation conditions on continental and global scales using the Advanced Very High Resolution Radiometer (AVHRR) sensor onboard the NOAA series of satellites

NDVI Images

- NDVI image provides an indication of the vigor and density of vegetation at the surface
- NDVI Images are sometimes referred to as "greenness maps" since they represent the vegetative vigor of plants
- These images help compare the current state of vegetation with previous time periods, for example the same time in an average year or a reference year (a particularly good or bad year) to detect anomalous conditions
- NDVI values can also be calculated per administrative unit and/or per land cover or crop cover type

NDVI Images

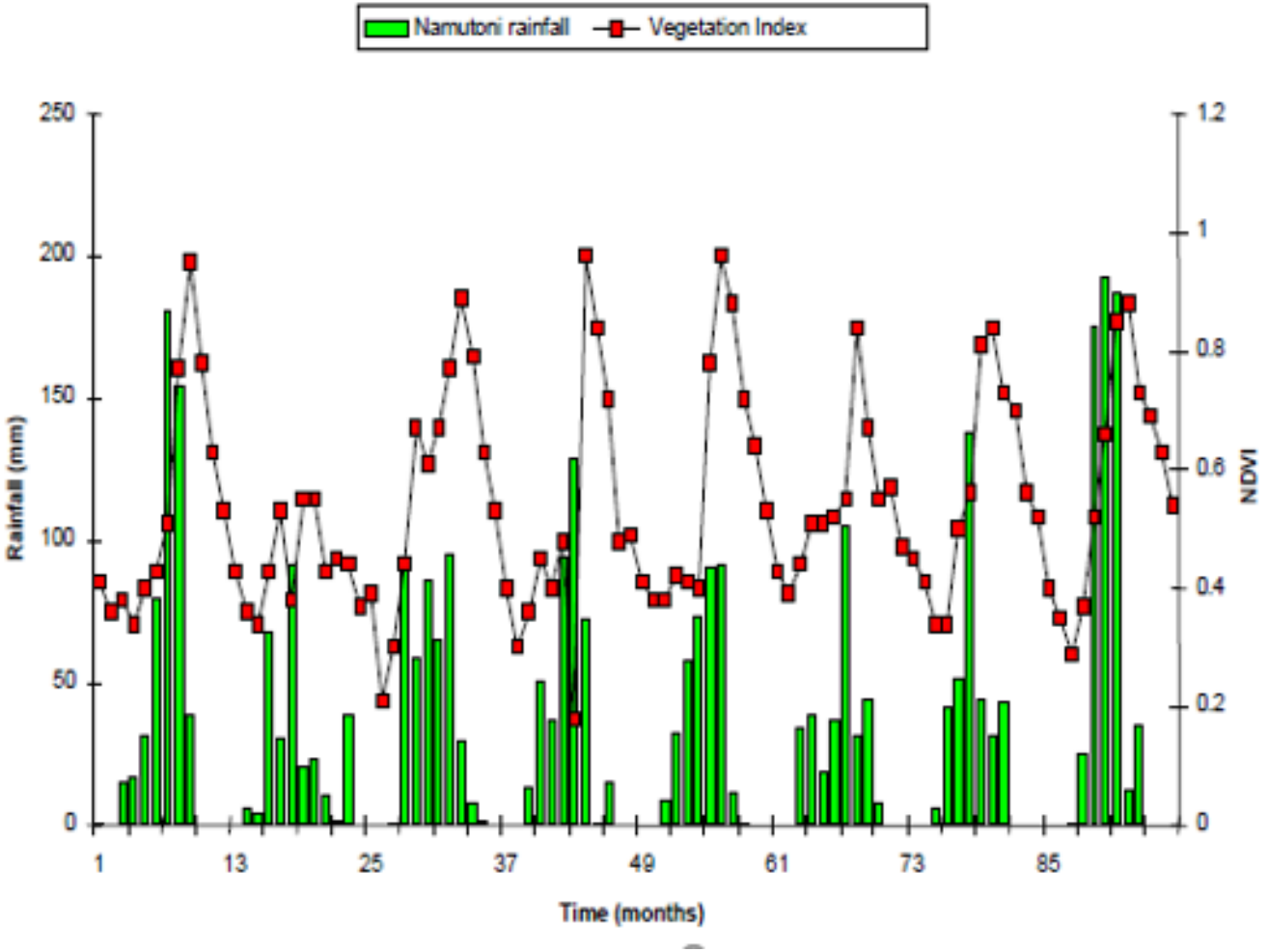
- In a cropped area, NDVI values steadily increase as the season progresses (from young green healthy crops), this reaches a maximum value before falling with senescence of the plant then harvest.
- The time series analysis of NDVI data provides a method of
  - Estimating net primary production over varying biomass types
  - Monitoring phenological patterns
  - Assessing the length of the crop growing season



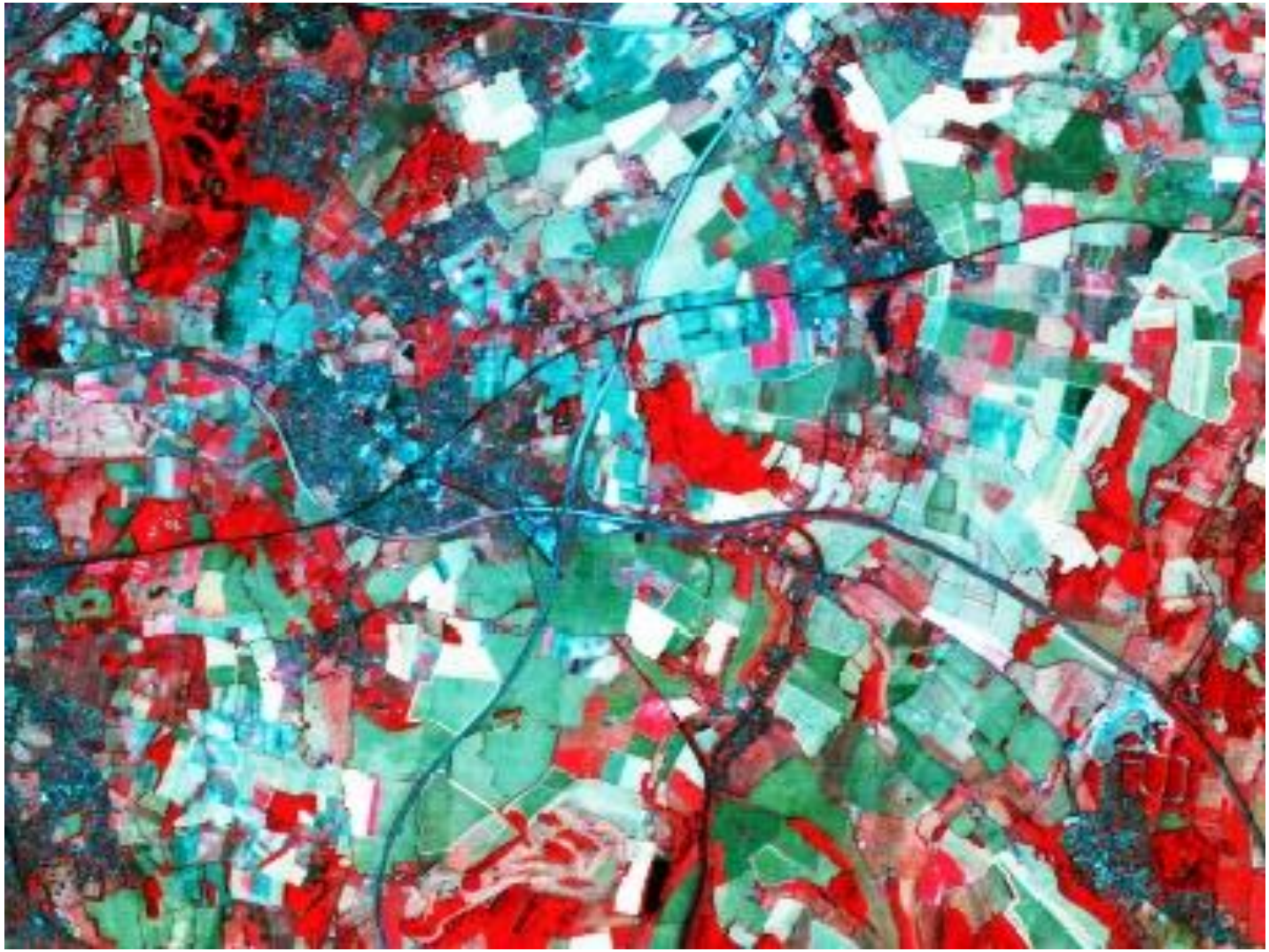
NDVI as a function of vegetation growth

Vegetation Indices -Rainfall

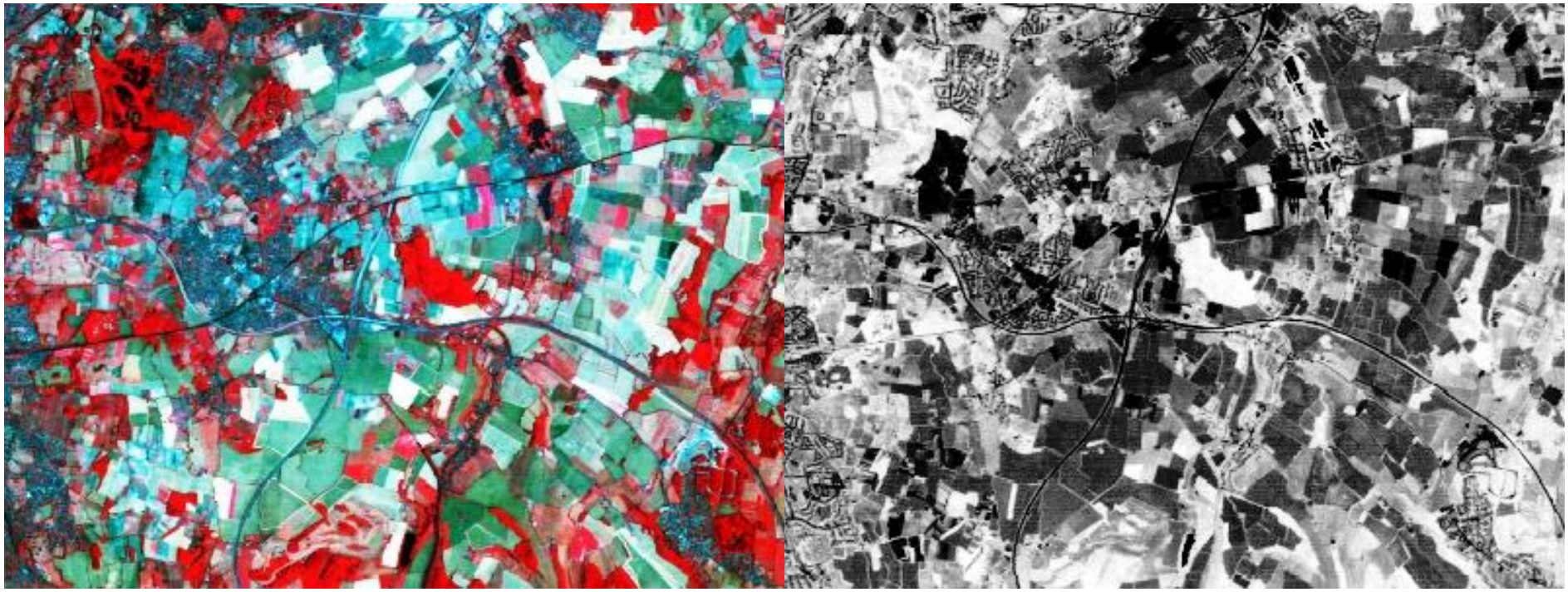
Monthly Rainfall and vegetation Response



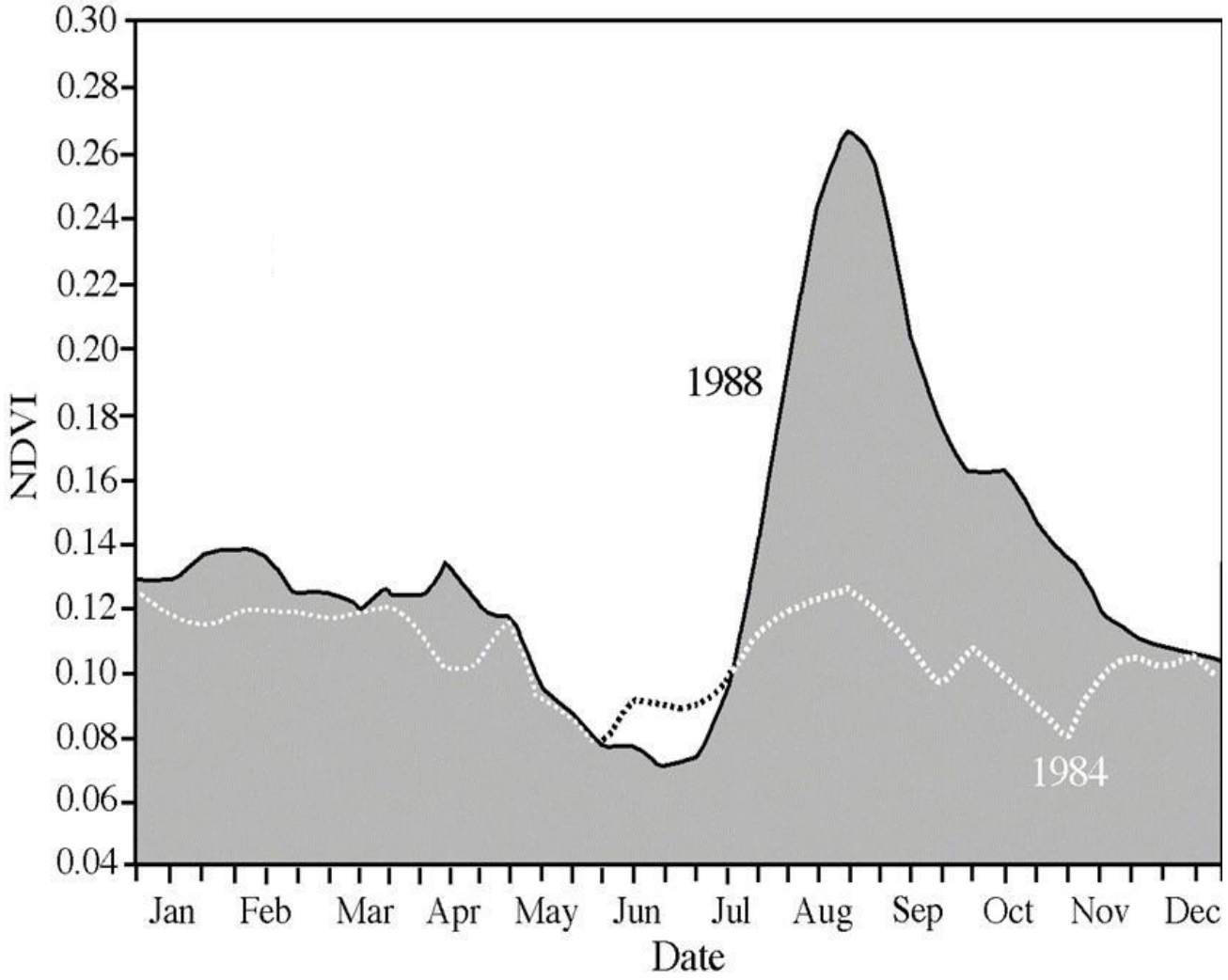
Visualization/analysis



NDVI

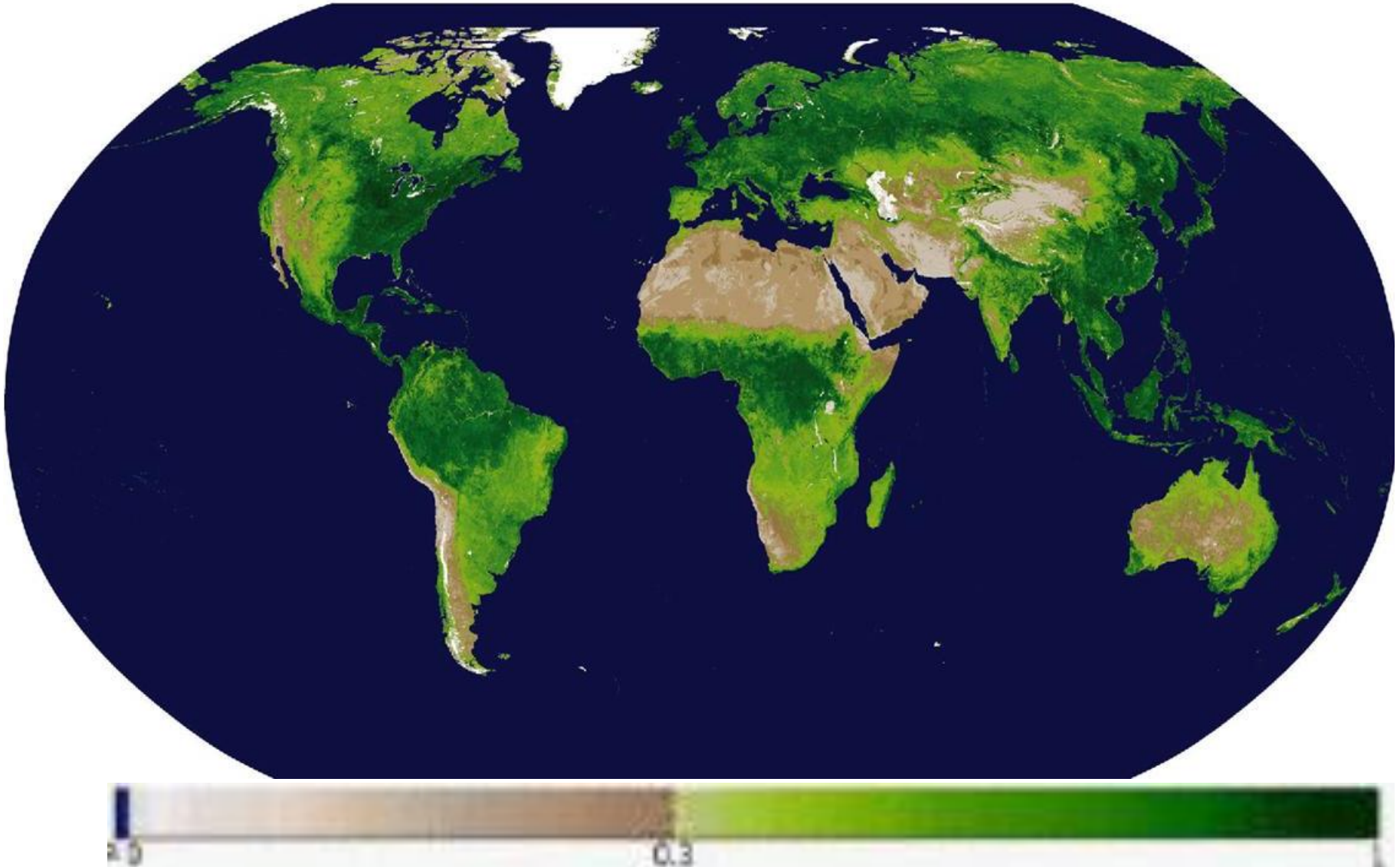


Time Series NDVI Measurements and Analysis

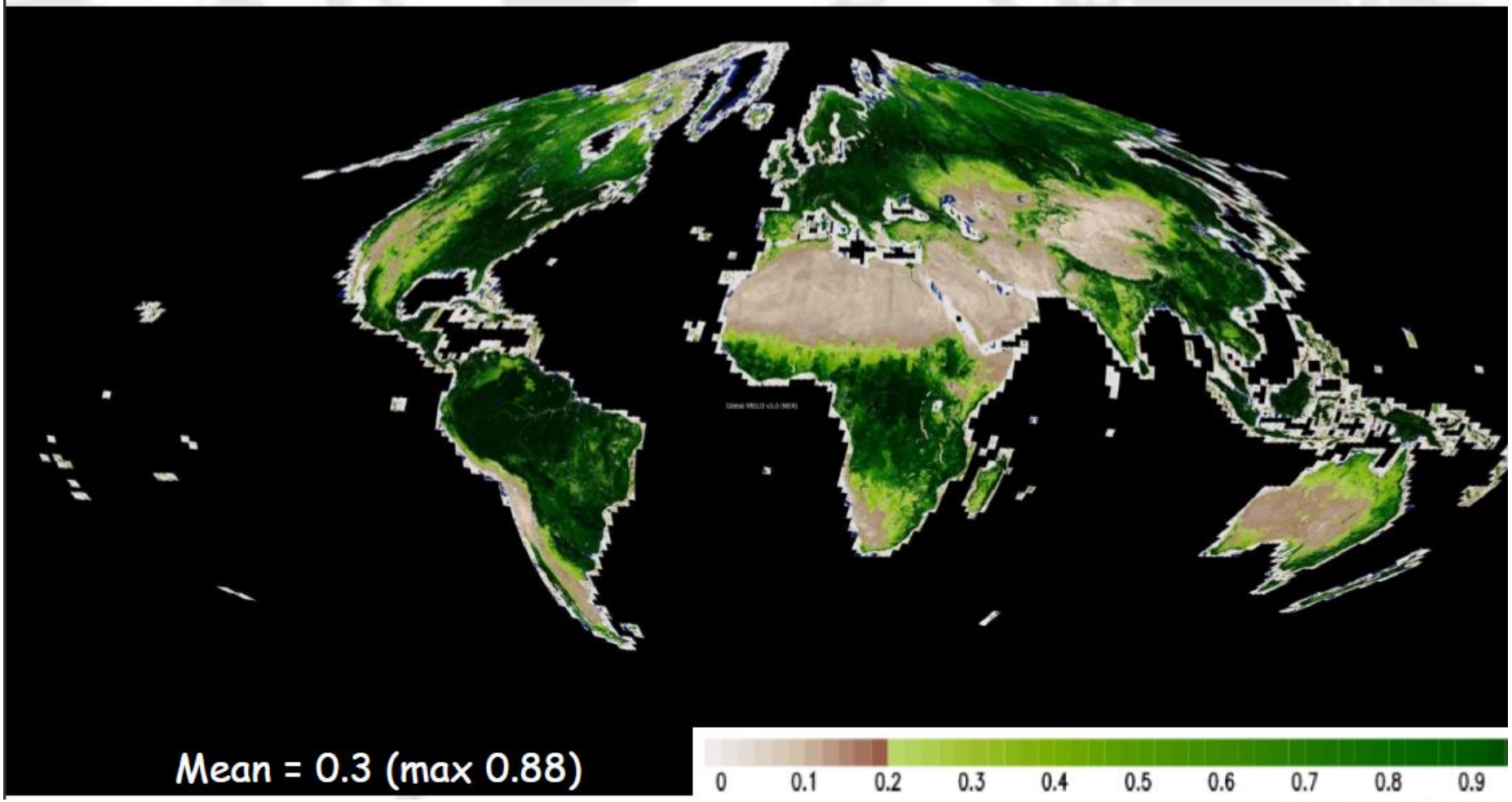


Time Series of 1984 and 1988 NDVI Measurements Derived from AVHRR Global Area Data

MODIS Vegetation Index Map of the World



Global annual 30 m Landsat NDVI



USGS
science for a changing world

NDVI

- Simple to use
- Require Red and NIR reflectance data (bands)
- Is used extensively to monitor vegetation
- Appears to be a poor indicator of vegetation biomass if the ground cover is low, as in arid and semi-arid regions

?